

Madelyn J Warr

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Education

Bachelor of Science, Mathematics.

Minors in Computer Science and Biology

University of Utah

GPA: 4.0

Expected May 2027

Salt Lake City, UT

Research Experience

Undergraduate Computer Science Researcher

University of Utah

August 2024-Present

Salt Lake City, UT

Advisor: Dr. Blair Sullivan, Professor

- Designed a fixed-parameter tractable algorithm which achieves a fairness objective in a graph and proved the problem is NP-hard
- Utilized high performance computing to run experiments on social networks with tens of thousands of nodes to verify the scalability of fairness heuristics for a manuscript which is under review
- Planned an experiment around a question relating to information fairness, collected and analyzed data, and presented findings to external collaborators

Research Experience for Undergraduates (NSF-funded)

Montana State University

June 2025-February 2026

Bozeman, MT

Advisor: Dr. Brendan Mumey, Professor

- Built a pipeline which processes biological data, predicts accessible chromatin regions with co-linear chaining and machine learning, and evaluates the results

Pre-Research Program in Mathematics

University of Utah

July 2024-August 2024

Salt Lake City, UT

Advisor: Kurt Vinhage, Assistant Professor

- Reasoned through complex mathematical problems related to dynamical systems with other undergraduates, professors, and graduate students
- Gained experience with LaTeX and mathematical proof-writing in a simulated mathematical research environment

Manuscripts

Dennis Robert Windham, Caroline J. Wendt, Alex Crane, **Madelyn J Warr**, Freda Shi, Sorelle A. Friedler, Blair D. Sullivan, and Aaron Clauset. Fast algorithms to improve fair information access in networks. *Accepted: PLOS Journal*. ArXiv:2409.03127

Madelyn Warr, Trung Dinh, Bella Root, Elyse Onstott, Kevin Yu, Joann Mudge, Thiruvarangan Ramaraj, Indika Kahanda, Brendan Mumey. PanACRpred: Predicting Accessible Chromatin Regions in Pangenomes using Motif Chaining. *Accepted: BICOB-2026 Conference Proceedings*. bioRxiv:2026.02.05.703812

Presentations

International Conference on Bioinformatics and Computational Biology March 2026
Honolulu, HI

Undergraduate Research Poster Session, Montana State University July 2025
Bozeman, MT

Skills

Languages

Java, Python, C#, SQL

General Tools

Linux, Command Line, Git, VS Code, IntelliJ, Jupyter Notebook, Pandas, HPC, SLURM

Bioinformatics Tools

Bedtools, BLAST, Integrative Genomic Viewer, MEME Suite, BioPython

Other

LaTeX, Unit Testing

Teaching Experience

STEM Camp Programming Instructor June 2024-July 2024
University of Utah Salt Lake City, UT

- Instructed over 30 children on the basics of computer programming
- Designed engaging and instructive projects and lessons

Relevant Coursework

Intro to Graph Theory: Graduate level

- Matchings, planarity, colorings, flows, extremal graph theory, Ramsey numbers, Hamiltonicity, tree-width
- Proof-writing, parameterized algorithms, probabilistic method

Discrete Mathematics

- Intro to proof writing, counting, permutations, set theory, functions

Algorithms

- Graph algorithms, dynamic programming, data structures, divide-and-conquer, computational complexity

Biochemistry

- Mechanisms for biological reactions, proteins, carbohydrates, nucleic acids, glycolysis / gluconeogenesis

Honors and Awards

Undergraduate Research Opportunity Program Scholar

December 2025

- Selected by the University of Utah to receive research funding

University of Utah Merit Scholarship

August 2023

Top 10 Award

June 2023

- Ranked in top 10 students in high school class of 800 based on GPA and course rigor

Young Technology Scholar Award

May 2023

- Selected by the Utah Board of Education for proficiency in information technology

Extracurricular / Volunteer Work

Undergraduate Involvement Co-chair

August 2025-Present

Association for Women in Mathematics

Salt Lake City, UT

- Planned engaging events to cultivate community in the University of Utah mathematics department, facilitate professional connections, and provide additional educational opportunities

Volunteer

May 2024-Present

Volunteers of America Youth Resource Center

Salt Lake City, UT

- Served food approximately once per month

Participant in Programming Competitions

October 2023, 2024, 2025

Salt Lake City, UT

Hosted by Lucid Software

Computer Science Ambassador

August 2022-May 2023

Jordan Academy for Technology and Careers

West Jordan, UT

- Prepared activities and delivered presentations to students in middle school to encourage the study of computer science